Gaussian Wasserstein Barycenters and the Raw Covariance Map

Fixed points, algorithms, contraction on compact spectral sets, and entropic and unbalanced extensions

Technical study

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Abstract

For positive weights λ_s summing to one and positive definite covariance matrices Σ_s, the centered Gaussian 2-Wasserstein barycenter has the unique covariance $\Sigma_\star$ satisfying

$$\Sigma_\star = \Psi_\lambda(\Sigma_\star), \qquad \Psi_\lambda(X) := \sum_s \lambda_s \big(X^{1/2}\Sigma_s X^{1/2}\big)^{1/2}.$$

The fixed-point identity is classical, but it does not by itself justify the raw Picard iteration $X_{n+1} = \Psi_\lambda(X_n)$. This note separates that raw map from the globally convergent transport/gradient update

$$K(X) = X^{-1/2}\Psi_\lambda(X)^2 X^{-1/2}.$$

It proves exact scalar and commuting results, derives the full Fréchet derivative of Ψ_λ through two Sylvester equations, gives an explicit sufficient contraction certificate on compact spectral boxes, and formulates the remaining noncommutative issue as a precise derivative-product problem. In particular, on each fixed invariant scalar interval bounded away from zero, a sufficiently high iterate Ψ_λ^k is a contraction; nevertheless, no fixed k works uniformly over all such intervals or all condition numbers. Thus a common one-dimensional objection has the wrong quantifier, while a general matrix contraction theorem remains unsupported by the literature located for this study.

The note also proves Gaussian preservation of ordinary Wasserstein barycenters from Gelbrich's nonexpansive Gaussianization map, distinguishes three inequivalent notions of entropic/Sinkhorn barycenter, and explains why the same Gaussianization argument fails for ambient Hellinger–Kantorovich/Wasserstein–Fisher–Rao geometry. At the Hellinger endpoint the barycenter of Gaussian densities is generically a Gaussian mixture, not a Gaussian. A Gaussian-restricted BWFR geometry is presented as a valid but explicitly constrained alternative.

Contents

1 Executive conclusions

The conclusions are easiest to state before the details.

1. A compact subset of strictly positive definite matrices must be bounded away from the boundary. The natural model domain is

$$\mathcal{K}_{m,M} := \{X \in \mathbb{S}^d_{++} : mI_d \preceq X \preceq MI_d\}, \qquad 0 < m \le M < \infty.$$

   Saying that a map is "contractive" is incomplete until a metric, a domain, and whether the map preserves that domain have been specified.

2. The Gaussian covariance equation has a unique positive definite solution. This follows from the Wasserstein barycenter theory of Agueh–Carlier and from matrix-analytic convexity arguments [5, 9]. Brouwer's theorem proves existence of a fixed point on a compact spectral box; it does not prove that the raw Picard iterates converge.

3. The algorithm with a classical global convergence theorem is

$$X_{n+1} = K(X_n) := X_n^{-1/2}\Psi_\lambda(X_n)^2 X_n^{-1/2},$$

   not $X_{n+1} = \Psi_\lambda(X_n)$. The two maps have the same positive definite fixed point but different dynamics. In one dimension K reaches the answer in one step, whereas the raw map only halves the exponent at each iteration.

4. In dimension one, with $c := \sum_s \lambda_s \sqrt{\Sigma_s}$ and $x_\star = c^2$,

$$\Psi(x) = c\sqrt{x}, \qquad \Psi^k(x) = x_\star^{1-2^{-k}} x^{2^{-k}}.$$

   On an invariant interval $[m, M]$ the exact Euclidean contraction constant is

$$q_k = 2^{-k}\left(\frac{x_\star}{m}\right)^{1-2^{-k}}.$$

   Hence $q_k \to 0$ for every fixed $m > 0$, but for every prescribed k one can choose m so small that $q_k > 1$. The correct statement is therefore: eventual contraction holds on each fixed compact interval, but there is no universal iterate independent of the lower spectral bound.

5. For general noncommuting matrices, this note gives (i) the exact derivative, (ii) a computable local spectral-radius criterion, and (iii) a conservative analytic compact-box certificate. These yield rigorous contraction results in several regimes, including commuting data and sufficiently well-conditioned boxes. A theorem covering every collection of positive definite inputs on every invariant compact spectral box was not found in the primary literature. The older numerical reports concerning the raw map are not rigorous counterexamples.

6. Gaussian Wasserstein barycenters are Gaussian because moment-matching Gaussianization is 1-Lipschitz for W_2 by Gelbrich's inequality. This is a nonexpansive projection argument, not a strict contraction argument.

7. “Entropic barycenter” can mean a product-reference entropic OT barycenter, a Schrodinger barycenter with Lebesgue/Brownian reference, or a debiased Sinkhorn-divergence barycenter. They have different covariance equations and opposite variance biases. General contraction of their covariance Picard maps is not known; scalar derivatives already rule out a generic one-step Euclidean contraction near the positive-semidefinite boundary.
8. Ordinary ambient HK/WFR barycenters do not admit a Gelbrich-style Gaussianization proof. Moment Gaussianization is not even continuous for HK on a compact covariance range. At the pure Hellinger endpoint, the exact barycenter of Gaussian-shaped densities is generically a finite Gaussian mixture. Gaussian-restricted BWFR barycenters are meaningful finite- dimensional surrogates, but Gaussianity then holds by restriction, not by an ambient preservation theorem.

2 Problem and notation

Let $\Sigma_1, \dots, \Sigma_N \in \mathbb{S}^d_{++}$ and let $\lambda_s > 0$ with $\sum_{s=1}^N \lambda_s = 1$. On $\mathbb{S}^d_+$ define the Bures–Wasserstein distance

$$d^2_{\mathrm{BW}}(A, B) := \operatorname{tr} A + \operatorname{tr} B - 2\operatorname{tr}\big(A^{1/2} B A^{1/2}\big)^{1/2}. \tag{1}$$

For Gaussian laws,

$$W_2^2(\mathcal{N}(m, A), \mathcal{N}(n, B)) = \|m - n\|_2^2 + d^2_{\mathrm{BW}}(A, B). \tag{2}$$

The covariance part of the barycenter problem is

$$\min_{X \in \mathbb{S}^d_+} F(X), \qquad F(X) := \frac{1}{2}\sum_{s=1}^N \lambda_s d^2_{\mathrm{BW}}(X, \Sigma_s). \tag{3}$$

The map under study is

$$\boxed{\Psi(X) = \Psi_\lambda(X) := \sum_{s=1}^N \lambda_s R_s(X), \qquad R_s(X) := \big(X^{1/2}\Sigma_s X^{1/2}\big)^{1/2}.} \tag{4}$$

It is continuous and smooth on $\mathbb{S}^d_{++}$, maps $\mathbb{S}^d_{++}$ into $\mathbb{S}^d_{++}$, and is positively homogeneous of degree 1/2:

$$\Psi(tX) = t^{1/2}\Psi(X), \qquad t > 0. \tag{5}$$

It is orthogonally equivariant, but it is not generally equivariant under an arbitrary congruence. That loss of full congruence equivariance is one source of difficulty in the noncommutative contraction problem.

We use $\|\cdot\|_F$ for the Frobenius norm and $\|\cdot\|_{\mathrm{op}}$ for the spectral norm. Unless explicitly stated otherwise, “contraction” below means a strict Lipschitz contraction for $\|\cdot\|_F$.

Remark 2.1 (Positive definite versus positive semidefinite)**.** The phrase “a compact domain of positive PSD matrices” should be made precise. A compact subset of $\mathbb{S}^d_{++}$ has a uniform lower eigenvalue bound $m > 0$. If singular matrices are admitted and the domain meets the boundary, the scalar square root is not Lipschitz there. Thus no Euclidean Lipschitz theorem uniform down to 0 can hold even in dimension one.

3 Why Wasserstein barycenters of Gaussians are Gaussian

3.1 Gelbrich's inequality

For a probability measure $\mu \in \mathcal{P}_2(\mathbb{R}^d)$, write m_μ and C_μ for its mean and covariance. Its moment-matching Gaussianization is

$$\mathcal{G}(\mu) := \mathcal{N}(m_\mu, C_\mu). \tag{6}$$

Theorem 3.1 (Gelbrich bound)**.** *For $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$W_2^2(\mu,\nu) \geq \|m_\mu - m_\nu\|_2^2 + d_{\mathrm{BW}}^2(C_\mu, C_\nu). \tag{7}$$

Equality holds for Gaussian measures and, more generally, for compatible location-scatter families. Consequently,

$$W_2(\mathcal{G}(\mu), \mathcal{G}(\nu)) \leq W_2(\mu,\nu). \tag{8}$$

Proof. Take any coupling (X,Y) of μ and ν and center it: $\widetilde{X} = X - m_\mu$, $\widetilde{Y} = Y - m_\nu$. If $C = \mathbb{E}[\widetilde{X}\widetilde{Y}^\top]$, then

$$\mathbb{E}\,\|X - Y\|_2^2 = \|m_\mu - m_\nu\|_2^2 + \operatorname{tr} C_\mu + \operatorname{tr} C_\nu - 2\operatorname{tr} C. \tag{9}$$

The block covariance is positive semidefinite:

$$\begin{pmatrix} C_\mu & C \\ C^\top & C_\nu \end{pmatrix} \succeq 0.$$

When the covariances are invertible, the block factorization criterion gives $C = C_\mu^{1/2} K C_\nu^{1/2}$ for a contraction K; the singular case follows by approximation. Von Neumann's trace inequality, or equivalently the polar factor maximization, yields

$$\operatorname{tr} C \leq \operatorname{tr}(C_\mu^{1/2} C_\nu C_\mu^{1/2})^{1/2}.$$

Substitution in (9) and minimization over couplings give (7). For two Gaussians, the maximizing cross-covariance is realized by a jointly Gaussian coupling, which proves equality and (8). □

This result is due to Gelbrich [1]; the Gaussian distance formula was obtained earlier in [2, 3, 4].

3.2 Gaussian preservation of the barycenter

Theorem 3.2 (Gaussian barycenter by Gaussianization)**.** *Let $\nu_s = \mathcal{N}(m_s, \Sigma_s)$ with $\Sigma_s \succ 0$. The unique minimizer of*

$$\mu \longmapsto \frac{1}{2}\sum_s \lambda_s W_2^2(\mu, \nu_s) \tag{10}$$

over $\mathcal{P}_2(\mathbb{R}^d)$ is Gaussian. Its mean is $m_\star = \sum_s \lambda_s m_s$, and its covariance is the unique minimizer of (3).

Proof. Since $\mathcal{G}(\nu_s) = \nu_s$, theorem 3.1 gives, for every candidate μ,

$$W_2(\mathcal{G}(\mu), \nu_s) \le W_2(\mu, \nu_s) \quad \text{for every } s.$$

Thus Gaussianization cannot increase the barycenter objective. General Wasserstein barycenter theory gives uniqueness when at least one input is absolutely continuous; here every nondegenerate Gaussian is absolutely continuous. Hence the unique minimizer equals its Gaussianization. Formula (2) separates mean and covariance, and the mean minimizer is the weighted Euclidean mean. □

Remark 3.3. The Gaussianization map is nonexpansive, not strictly contractive. Equality holds for every pair of Gaussians. Gaussian preservation should therefore not be confused with a Banach contraction mechanism.

4 The covariance equation and its unique solution

4.1 Optimal maps and first-order condition

The optimal linear transport from $\mathcal{N}(0, X)$ to $\mathcal{N}(0, \Sigma_s)$ is

$$T_s(X) := X^{-1/2} R_s(X) X^{-1/2}. \tag{11}$$

It is symmetric positive definite and satisfies $T_s(X) X T_s(X) = \Sigma_s$. Define its weighted mean

$$H(X) := \sum_s \lambda_s T_s(X). \tag{12}$$

Since $R_s(X) = X^{1/2} T_s(X) X^{1/2}$,

$$\Psi(X) = X^{1/2} H(X) X^{1/2}. \tag{13}$$

Proposition 4.1 (Differential of the Bures objective)**.** *For a symmetric perturbation Z,*

$$D\, d^2_{\mathrm{BW}}(X, \Sigma_s)[Z] = \operatorname{tr}\big((I_d - T_s(X))Z\big), \tag{14}$$

and therefore

$$\nabla F(X) = \frac{1}{2}(I_d - H(X)). \tag{15}$$

Proof. The non-linear trace term may equivalently be written

$$\operatorname{tr} R_s(X) = \operatorname{tr}\big(\Sigma_s^{1/2} X \Sigma_s^{1/2}\big)^{1/2}.$$

For a positive definite Y, spectral calculus and cyclicity of the trace give

$$D\ \operatorname{tr}(Y^{1/2})[W] = \frac{1}{2} \operatorname{tr}(Y^{-1/2} W).$$

With $Y = \Sigma_s^{1/2} X \Sigma_s^{1/2}$, this yields

$$D\ \operatorname{tr} R_s(X)[Z] = \frac{1}{2} \operatorname{tr}\left(\Sigma_s^{1/2} (\Sigma_s^{1/2} X \Sigma_s^{1/2})^{-1/2} \Sigma_s^{1/2} Z\right) = \frac{1}{2} \operatorname{tr}(T_s(X) Z),$$

where the last matrix is the equivalent target-side formula for the Gaussian optimal map. Differentiating (1) proves (14); summing proves (15). □

Thus the stationarity equation is

$$H(X) = I_d. \tag{16}$$

By (13), this is equivalent to

$$\boxed{\Psi(X) = X.} \tag{17}$$

Theorem 4.2 (Existence and uniqueness)**.** *For $\Sigma_s \succ 0$ and positive weights summing to one, there is a unique $\Sigma_\star \succ 0$ satisfying* (17)*. It is the unique minimizer of* (3)*.*

This is the Gaussian part of Agueh–Carlier's barycenter theorem [5, Theorem 6.1]. Earlier work derived the equation and studied the associated multi-coupling problem [6, 7]. Bhatia–Jain–Lim give a matrix-analysis proof based on convexity of the fidelity functional [9].

For completeness, strict uniqueness can also be seen directly. The map $Y \mapsto \operatorname{tr}(Y^{1/2})$ is strictly concave on positive definite matrices. As $X \mapsto \Sigma_s^{1/2} X \Sigma_s^{1/2}$ is injective when $\Sigma_s \succ 0$, each function

$$X \longmapsto \operatorname{tr} X + \operatorname{tr} \Sigma_s - 2 \operatorname{tr}(\Sigma_s^{1/2} X \Sigma_s^{1/2})^{1/2}$$

is strictly convex. Hence F has at most one stationary point, and the fixed point constructed on the compact box below is that unique minimizer.

4.2 The compact spectral box used for existence

Assume

$$a_s I_d \preceq \Sigma_s \preceq b_s I_d, \qquad 0 < a_s \leq b_s. \tag{18}$$

Set

$$c_- := \sum_s \lambda_s \sqrt{a_s}, \qquad c_+ := \sum_s \lambda_s \sqrt{b_s}. \tag{19}$$

Lemma 4.3 (Spectral enclosure)**.** *If $X \in \mathcal{K}_{m,M}$, then*

$$c_- \sqrt{m}\, I_d \preceq \Psi(X) \preceq c_+ \sqrt{M}\, I_d. \tag{20}$$

Consequently, $\mathcal{K}_{m,M}$ is forward invariant under Ψ whenever

$$m \leq c_-^2, \qquad M \geq c_+^2. \tag{21}$$

Proof. From (18),

$$a_s X \preceq X^{1/2} \Sigma_s X^{1/2} \preceq b_s X.$$

The matrix square root is operator monotone, so for $X \in \mathcal{K}_{m,M}$,

$$\sqrt{a_s m}\, I_d \preceq R_s(X) \preceq \sqrt{b_s M}\, I_d.$$

Sum with weights. Conditions (21) are exactly $c_- \sqrt{m} \geq m$ and $c_+ \sqrt{M} \leq M$. □

Thus Ψ is a continuous self-map of a convex compact set, and Brouwer gives a fixed point. This is an existence argument only. A continuous self-map of a compact convex set need not be a contraction, and its Picard iterates need not converge.

5 Algorithms in the literature: the essential distinction

5.1 Raw Picard versus the transport update

The raw fixed-point iteration is

$$X_{n+1} = \Psi(X_n). \tag{22}$$

The measure pushforward construction of Alvarez-Esteban et al. instead gives the covariance update

$$\boxed{K(X) := H(X)XH(X) = X^{-1/2}\Psi(X)^2X^{-1/2}.} \tag{23}$$

The identity follows from (13). Both maps have the same positive definite fixed point, but they are not the same map.

More generally, a damped Bures–Wasserstein gradient step has the form

$$K_\eta(X) = ((1-\eta)I_d + \eta H(X))X((1-\eta)I_d + \eta H(X)), \tag{24}$$

with normalization depending on whether the objective contains a factor 1/2. The unit step $\eta = 1$ is (23).

Theorem 5.1 (Global convergence of the stabilized update)**.** *Suppose the input covariances are positive semidefinite and at least one is positive definite. Starting from $X_0 \succ 0$, the sequence $X_{n+1} = K(X_n)$ converges to the Bures–Wasserstein barycenter covariance. Moreover, the barycenter variance decreases and controls the step:*

$$F(X) \geq F(K(X)) + \frac{1}{2}d^2_{\mathrm{BW}}(X, K(X)), \tag{25}$$

up to the normalization of F.

This is the covariance specialization of [8, Theorem 4.2]; see also [9]. Determinant control keeps the iterates away from the boundary, the variance inequality identifies cluster points as fixed points, and uniqueness finishes the proof.

Example 5.2 (The scalar contrast)**.** In dimension one, $H(x) = c/\sqrt{x}$ and therefore

$$K(x) = \frac{c}{\sqrt{x}}x\frac{c}{\sqrt{x}} = c^2 = x_\star.$$

The stabilized update is exact in one step. The raw map is $x \mapsto c\sqrt{x}$ and generally needs infinitely many steps.

5.2 What is known about the raw iteration

Ruschendorf–Uckelmann considered an equivalent square-root iteration in the multi-coupling formulation [7]. Alvarez-Esteban et al. explicitly distinguish it from (23), label it their iteration (27), and state that no theoretical consistency result was available for it [8, Section 5]. They report satisfactory two-dimensional examples and quote unfavorable behavior for some three-dimensional initializations. These are numerical observations, not a rigorous divergent orbit or cycle.

As of the literature audit for this note, no primary theorem was located that proves a Banach contraction of the raw map Ψ, or of a fixed iterate Ψ^k, on every invariant spectral compact. Later global and linear-rate results concern K or damped Riemannian gradient descent, not raw Picard:

- Chewi–Maunu–Rigollet–Stromme prove global rates for Bures gradient descent and stochastic gradient descent using a Polyak–Lojasiewicz inequality [11].
- Altschuler–Chewi–Gerber–Stromme obtain dimension-free rates under uniform eigenvalue bounds. For a damped step and condition number κ, the objective gap decays like $\exp(-cT/\kappa^{5/2})$ with a universal constant $c > 0$ [12].
- Zemel–Panaretos connect the update with Procrustes alignment and Wasserstein gradient descent [10].
- Brahmachari–Rubboli–Tomamichel prove exponential convergence for a broader Bures projection fixed-point algorithm that specializes to the Alvarez-Esteban K update [13].

5.3 Other computational formulations

The Bures fidelity has the semidefinite representation

$$\operatorname{tr}\big(X^{1/2}\Sigma_s X^{1/2}\big)^{1/2} = \max_{Z_s} \left\{ \operatorname{tr} Z_s : \begin{pmatrix} X & Z_s \\ Z_s^\top & \Sigma_s \end{pmatrix} \succeq 0 \right\}. \tag{26}$$

Consequently, (3) is an SDP:

$$\begin{aligned} \min_{X, Z_1, \ldots, Z_N} \quad & \sum_s \lambda_s \big(\operatorname{tr} X + \operatorname{tr} \Sigma_s - 2 \operatorname{tr} Z_s\big), \\ \text{subject to} \quad & \begin{pmatrix} X & Z_s \\ Z_s^\top & \Sigma_s \end{pmatrix} \succeq 0, \qquad s = 1, \ldots, N. \end{aligned} \tag{27}$$

It is robust but expensive at large d or N. Other choices include Euclidean projected gradient, Bures Riemannian gradient or trust-region methods, stochastic gradient for population barycenters, and generalized Procrustes alternating alignment. For the Gaussian covariance problem, the stabilized update (23) remains especially simple: each step requires matrix square roots and linear solves but no step-size selection.

6 What a contraction statement must say

Definition 6.1. Let (D, d) be a metric space. A map $G : D \to D$ is a strict contraction if there is $q < 1$ such that

$$d(G(X), G(Y)) \le q d(X, Y), \qquad X, Y \in D.$$

An iterate G^k is an eventual contraction if this holds for some finite k.

Three quantifiers are distinct:

1. for each fixed data set and each fixed invariant compact D, there exists a $k = k(D, \{\Sigma_s\})$;
2. for each fixed data set, one k works for every invariant compact;
3. one universal k works for every data set and compact.

The scalar calculation below proves the first and disproves the second and third in their unrestricted form.

Metric choice matters. The Frobenius, operator, affine-invariant, log-Euclidean, Thompson, and Bures metrics can give different Lipschitz constants. Equivalent norms on a finite-dimensional compact have the same topology, but "constant < 1" is not invariant under a change of norm.

Proposition 6.2 (Why eventual contraction would settle raw Picard)**.** *Let D be complete and forward invariant under a continuous map G. If $G^k : D \to D$ is a contraction for some k, then G has a unique fixed point p in D, and every raw orbit $G^n(X)$ converges to p.*

Proof. Banach's theorem gives a unique fixed point p of G^k and convergence of $G^{kn}(X)$ to p. Since $G(p)$ is also fixed by G^k,

$$G^k(G(p)) = G(G^k(p)) = G(p),$$

uniqueness implies $G(p) = p$. Continuity then gives $G^{kn+r}(X) \to G^r(p) = p$ for each $r = 0, \dots, k-1$. □

Thus a rigorous nonconvergent raw orbit inside an invariant compact would disprove eventual contraction there. The historical numerical remarks do not yet provide such a mathematical counterexample.

7 Exact scalar, isotropic, and commuting analysis

7.1 Dimension one

Write the scalar covariances as $\sigma_s > 0$ and define

$$c := \sum_s \lambda_s \sqrt{\sigma_s}, \qquad x_\star := c^2. \tag{28}$$

Then $\Psi(x) = c\sqrt{x}$.

Theorem 7.1 (Exact iterates and exact Euclidean constant)**.** *For every $k \geq 1$,*

$$\Psi^k(x) = x_\star^{1-2^{-k}} x^{2^{-k}}, \tag{29}$$

and on $[m, M] \subset (0, \infty)$,

$$\mathrm{Lip}(\Psi^k; [m, M]) = q_k := 2^{-k} \left(\frac{x_\star}{m}\right)^{1-2^{-k}}. \tag{30}$$

The interval is forward invariant exactly when $m \leq x_\star \leq M$. On every such fixed interval, $q_k \to 0$, so Ψ^k is a contraction for all sufficiently large k.

Proof. The identity is true for $k = 1$. If it holds with exponent $p = 2^{-k}$, then

$$\Psi^{k+1}(x) = \sqrt{x_\star}\left(x_\star^{1-p} x^p\right)^{1/2} = x_\star^{1-p/2} x^{p/2}.$$

This proves (29). Its derivative is

$$(\Psi^k)'(x) = 2^{-k} x_\star^{1-2^{-k}} x^{2^{-k}-1}.$$

It is positive and decreasing, so its maximum on $[m, M]$ is attained at m, which gives (30). Since Ψ is increasing, invariance is equivalent to $\Psi(m) \geq m$ and $\Psi(M) \leq M$, or $m \leq c^2 \leq M$. Finally $2^{-k} \to 0$ while $(x_\star/m)^{1-2^{-k}} \to x_\star/m$, hence $q_k \to 0$. □

Corollary 7.2 (No uniform iterate)**.** *For every prescribed k, there is an invariant compact interval on which Ψ^k is not a Euclidean contraction.*

Proof. Fix $x_\star$ and $M \geq x_\star$. Formula (30) exceeds one when $m/x_\star$ is sufficiently small, while $m \leq x_\star$ keeps the interval invariant. □

For $k = 1$, the constant is

$$q_1 = \frac{1}{2}\sqrt{\frac{x_\star}{m}}, \tag{31}$$

so raw Picard expands near the lower end whenever $m < x_\star/4$.

In contrast, the scalar Thompson/log metric gives a global exact contraction:

$$|\log \Psi(x) - \log \Psi(y)| = \frac{1}{2}\,|\log x - \log y|\,. \tag{32}$$

This illustrates why the metric cannot be omitted from the claim.

7.2 Isotropic data in arbitrary dimension

Suppose $\Sigma_s = \sigma_s I_d$. Then

$$\Psi(X) = cX^{1/2}, \qquad \Psi^k(X) = x_\star^{1-2^{-k}} X^{2^{-k}}. \tag{33}$$

Proposition 7.3 (Exact isotropic compact-box constant)**.** *On $\mathcal{K}_{m,M}$,*

$$\mathrm{Lip}_F(\Psi^k; \mathcal{K}_{m,M}) = 2^{-k}\left(\frac{x_\star}{m}\right)^{1-2^{-k}}. \tag{34}$$

Thus the scalar conclusions, including nonexistence of a universal k, hold in every matrix dimension.

Proof. For $0 < p < 1$, the Fréchet derivative of $X \mapsto X^p$ in an eigenbasis has coefficients equal to divided differences of t^p. On $[m, M]$ their maximum is pm^{p-1} by concavity. This gives the upper bound in (34); the scalar direction at $X = mI_d$ attains it. □

7.3 Commuting inputs and commuting iterates

Suppose all inputs and the initial matrix commute. In a common orthogonal basis, write

$$\Sigma_s = \mathrm{diag}(\sigma_{s,1}, \ldots, \sigma_{s,d}), \qquad X = \mathrm{diag}(x_1, \ldots, x_d),$$

and

$$c_j := \sum_s \lambda_s \sqrt{\sigma_{s,j}}, \qquad x_{\star,j} = c_j^2. \tag{35}$$

Then the coordinates decouple:

$$(\Psi^k(X))_{jj} = x_{\star,j}^{1-2^{-k}} x_j^{2^{-k}}. \tag{36}$$

On a diagonal rectangle $m_j \leq x_j \leq M_j$, the exact Frobenius Lipschitz constant is

$$\max_j 2^{-k}\left(\frac{x_{\star,j}}{m_j}\right)^{1-2^{-k}}. \tag{37}$$

It tends to zero on every fixed rectangle bounded away from zero. In coordinatewise log distance the one-step factor is exactly $1/2$.

The preceding argument restricts both points to the commuting algebra. The next result studies arbitrary symmetric perturbations around the commuting barycenter and is stronger.

Theorem 7.4 (Local derivative for commuting data)**.** *Assume the Σ_s are diagonal in a common basis. At the barycenter $X_\star = \mathrm{diag}(c_1^2, \dots, c_d^2)$, the derivative $J = D\Psi(X_\star)$ is diagonal on the standard symmetric matrix basis. For a diagonal direction E_{ii} its eigenvalue is $1/2$. For an off-diagonal direction $E_{ij} + E_{ji}$, let $u_s = \sqrt{\sigma_{s,i}}$ and $v_s = \sqrt{\sigma_{s,j}}$. The eigenvalue is*

$$\theta_{ij} = \frac{1}{c_i + c_j} \sum_s \lambda_s \frac{c_i u_s^2 + c_j v_s^2}{c_i u_s + c_j v_s}. \tag{38}$$

It satisfies

$$\frac{1}{2} \leq \theta_{ij} < 1. \tag{39}$$

Hence $\|D\Psi(X_\star)\|_{F\to F} < 1$, and Ψ is a Frobenius contraction on some full, not merely diagonal, neighborhood of $X_\star$.

Proof. The derivative formula established in theorem 8.1 below preserves each coordinate matrix when all matrices at the base point are diagonal. Substitution of $X_\star^{1/2} = \mathrm{diag}(c_1, \dots, c_d)$ gives (38); the diagonal coefficient reduces to $\sum_s \lambda_s u_s/(2c_i) = 1/2$.

For the upper bound, pointwise in s,

$$\frac{c_i u_s^2 + c_j v_s^2}{c_i u_s + c_j v_s} < u_s + v_s,$$

because the difference after multiplication by the positive denominator is $(c_i + c_j)u_s v_s > 0$. Averaging gives a strict upper bound $c_i + c_j$, and division by $c_i + c_j$ gives $\theta_{ij} < 1$.

For the lower bound, weighted Cauchy–Schwarz gives

$$c_i u_s^2 + c_j v_s^2 \geq \frac{(c_i u_s + c_j v_s)^2}{c_i + c_j}.$$

Therefore

$$\theta_{ij} \geq \frac{c_i^2 + c_j^2}{(c_i + c_j)^2} \geq \frac{1}{2}.$$

The derivative is self-adjoint and diagonal in an orthonormal Frobenius basis, so its operator norm is its largest eigenvalue, strictly below one. Continuity of $D\Psi$ yields a neighborhood on which the derivative norm remains below one; the mean-value formula then gives a contraction. □

Example 7.5 (Slow off-diagonal modes)**.** For a single input $A = \mathrm{diag}(a_1, \dots, a_d)$, the fixed point is A and

$$D\Psi(A)[E_{ij} + E_{ji}] = \left(1 - \frac{\sqrt{a_i a_j}}{a_i + a_j}\right)(E_{ij} + E_{ji}). \tag{40}$$

For $i \neq j$, the coefficient equals $1/2$ when $a_i = a_j$ and approaches 1 as $a_i/a_j \to \infty$; diagonal modes separately have eigenvalue $1/2$. Thus even local raw Picard convergence can be arbitrarily slow with ill-conditioning; no condition-number-free strict factor is possible.

8 Full noncommutative differential calculus

8.1 Derivative through Sylvester equations

For $Z \succ 0$, define the Lyapunov/Sylvester operator

$$\mathcal{L}_Z(U) := ZU + UZ. \tag{41}$$

It is invertible on symmetric matrices. Differentiating $S^2 = X$ shows that the Fréchet derivative of the matrix square root is

$$D(X^{1/2})[H] = \mathcal{L}_{X^{1/2}}^{-1}(H). \tag{42}$$

Theorem 8.1 (Exact derivative of the raw map)**.** *Fix $X \succ 0$ and a symmetric perturbation H. Put*

$$S = X^{1/2}, \qquad R_s = (S\Sigma_s S)^{1/2}.$$

Let U and V_s be the unique symmetric solutions of

$$SU + US = H, \tag{43}$$

$$R_s V_s + V_s R_s = U\Sigma_s S + S\Sigma_s U. \tag{44}$$

Then

$$\boxed{D\Psi(X)[H] = \sum_s \lambda_s V_s.} \tag{45}$$

Proof. Equation (43) is the derivative of $S^2 = X$. For $Y_s = S\Sigma_s S$, the product rule gives

$$DY_s[H] = U\Sigma_s S + S\Sigma_s U.$$

Differentiating $R_s^2 = Y_s$ gives (44). Sum with the fixed weights. □

The formula is numerically stable: diagonalize the two positive matrices or solve the Sylvester equations directly. It avoids finite differences and allows exact evaluation of the Jacobian on the $d(d+1)/2$-dimensional symmetric space.

8.2 A conservative compact-box Lipschitz bound

Lemma 8.2 (Square-root derivative norm)**.** *For $Z \succ 0$,*

$$\left\|\mathcal{L}_Z^{-1}\right\|_{F\to F} = \frac{1}{2\lambda_{\min}(Z)}. \tag{46}$$

Proof. In an eigenbasis of $Z = \operatorname{diag}(z_1, \dots, z_d)$, $(\mathcal{L}_Z^{-1}H)_{ij} = H_{ij}/(z_i + z_j)$. The largest coefficient is $1/(2\min_i z_i)$. □

Proposition 8.3 (One-step analytic bound)**.** *Under (18), on $\mathcal{K}_{m,M}$,*

$$\operatorname{Lip}_F(\Psi; \mathcal{K}_{m,M}) \le L(m, M) := \frac{\sqrt{M}}{2m} \sum_s \lambda_s \frac{b_s}{\sqrt{a_s}}. \tag{47}$$

Proof. Let $\|H\|_F = 1$. From (43),

$$\|U\|_F \le \frac{1}{2\sqrt{m}}.$$

Also $\|S\|_{\mathrm{op}} \le \sqrt{M}$ and $\|\Sigma_s\|_{\mathrm{op}} \le b_s$, hence

$$\|U\Sigma_s S + S\Sigma_s U\|_F \le b_s\sqrt{\frac{M}{m}}.$$

Since $S\Sigma_s S \succeq a_s m I_d$, we have $R_s \succeq \sqrt{a_s m} I_d$. A second use of lemma 8.2 gives

$$\|V_s\|_F \le \frac{b_s\sqrt{M}}{2m\sqrt{a_s}}.$$

After summing, this bounds $\|D\Psi(X)\|_{F\to F}$ uniformly on the convex box. Integrating the derivative along the segment from X to Y proves (47). □

This bound is deliberately elementary and can be loose. Its value is that it is explicit and composable.

8.3 A rigorous sufficient theorem for an iterate

Starting from $m_0 = m$, $M_0 = M$, define

$$m_{j+1} = c_-\sqrt{m_j}, \qquad M_{j+1} = c_+\sqrt{M_j}. \tag{48}$$

Equivalently,

$$m_j = c_-^{2(1-2^{-j})} m^{2^{-j}}, \qquad M_j = c_+^{2(1-2^{-j})} M^{2^{-j}}. \tag{49}$$

By lemma 4.3, $\Psi^j(\mathcal{K}_{m,M}) \subset \mathcal{K}_{m_j,M_j}$.

Theorem 8.4 (Compact-box contraction certificate)**.** *Let*

$$Q_k := \prod_{j=0}^{k-1} L(m_j, M_j), \tag{50}$$

where L is defined in (47)*. Then*

$$\mathrm{Lip}_F(\Psi^k; \mathcal{K}_{m,M}) \le Q_k. \tag{51}$$

In particular, $Q_k < 1$ is a rigorous sufficient certificate that Ψ^k is a Frobenius contraction on the box. If the box is invariant by (21)*, Banach's theorem applies.*

Moreover, put

$$L_\infty := \frac{c_+}{2c_-^2}\sum_s \lambda_s \frac{b_s}{\sqrt{a_s}}. \tag{52}$$

If $L_\infty < 1$, then $Q_k \to 0$ and some iterate is a contraction on every fixed box $\mathcal{K}_{m,M}$.

Proof. Each factor in the derivative chain for Ψ^k is evaluated in the corresponding image box. Apply proposition 8.3 successively to obtain (51). From (49), $m_j \to c_-^2$ and $M_j \to c_+^2$, so $L(m_j, M_j) \to L_\infty$. If the limit is below one, all sufficiently late factors are bounded by some $q < 1$, and the finite product of early factors times q^{k-j_0} tends to zero. □

The condition $L_\infty < 1$ is sufficient, not necessary. Failure of this test says only that the elementary eigenvalue estimates lost too much information. The exact derivative product may still contract.

8.4 The sharp local criterion

Let $X_\star$ be the barycenter and $J = D\Psi(X_\star)$. Homogeneity gives a universal eigenpair:

$$J[X_\star] = \frac{1}{2}X_\star. \tag{53}$$

Indeed, differentiate $\Psi(tX_\star) = \sqrt{t}X_\star$ at $t = 1$.

Theorem 8.5 (Local eventual contraction criterion)**.** *Let G be a C^1 map on a finite-dimensional normed space with fixed point p, and let $J = DG(p)$.*

1. *If $\|J\| < 1$ in the chosen induced norm, then G is a contraction on a sufficiently small invariant ball around p.*
2. *If the spectral radius $\rho(J) < 1$, then for every fixed induced norm there is a k such that G^k is a contraction on a sufficiently small invariant ball around p.*
3. *If $\rho(J) \geq 1$, no iterate can be a strict local contraction in any norm.*

Applied to $G = \Psi$, the best possible local factor is at least $1/2$ because of (53).

Proof. The first statement follows from continuity of DG and the mean-value formula. If $\rho(J) < 1$, then $J^k \to 0$, hence $\left\|J^k\right\| < 1$ for some k; use the first statement for G^k. Conversely, a local contraction of G^k implies $\left\|J^k\right\| < 1$, so $\rho(J)^k = \rho(J^k) < 1$. □

8.5 What remains open for the raw map

The general noncommuting problem can now be stated without ambiguity.

> For fixed positive definite data and a fixed forward-invariant spectral box, is there always a finite k for which
>
> $$\sup_{X \in \mathcal{K}_{m,M}} \left\| D\Psi(\Psi^{k-1}X) \cdots D\Psi(X) \right\|_{F \to F} < 1?$$

The exact scalar and commuting results support a positive answer in those regimes. The compact-box theorem proves it under an explicit conditioning condition. The literature located for this study does not settle the fully noncommuting statement. A useful first local target is

$$\rho(D\Psi(X_\star)) < 1 \quad \text{for arbitrary noncommuting positive definite inputs.} \tag{54}$$

Even a proof of (54) would be local; extending it uniformly over a whole spectral box would require control of derivative products along every raw orbit.

9 Practical derivative test and reference algorithms

For a concrete data set, the following procedure separates provable facts from numerical evidence.

Step 1: compute the barycenter reliably. Use the stabilized update

$$X \leftarrow X^{-1/2}\Psi(X)^2 X^{-1/2}$$

until either the fixed residual $\|\Psi(X) - X\|_F \,/\, \|X\|_F$ or the Bures step is below tolerance. Symmetrize after floating-point products and compute square roots by an eigendecomposition with positive eigenvalue checks.

Step 2: assemble the exact Jacobian. Choose the orthonormal symmetric basis

$$E_{ii}, \qquad (E_{ij} + E_{ji})/\sqrt{2} \quad (i < j).$$

For each basis element solve (43) and (44); project the result onto the basis. This forms the $d(d+1)/2$ square matrix representing $D\Psi(X_\star)$.

Step 3: distinguish three diagnostics.

- $\rho(J) < 1$ certifies local eventual contraction.
- $\|J\|_2 < 1$ certifies local one-step Frobenius contraction.
- Neither diagnostic certifies contraction on a large compact box. For that, bound the derivative chain along the full reachable set, using theorem 8.4 or sharper interval/spectral estimates.

Step 4: search for a rigorous obstruction if needed. A pair X, Y with

$$\left\|\Psi^k(X) - \Psi^k(Y)\right\|_F \geq \|X - Y\|_F$$

disproves k-step Frobenius contraction on any domain containing them, but does not disprove eventual contraction for a larger iterate. A verified nontrivial periodic orbit inside an invariant compact would disprove every eventual-contraction theorem there.

10 Entropic and Sinkhorn Gaussian barycenters

10.1 Three objectives that must not be conflated

Fix the convention

$$\mathrm{OT}_\varepsilon(\mu, \nu) := \inf_{\pi \in \Pi(\mu,\nu)} \left\{ \frac{1}{2} \int \|x - y\|_2^2 \,\mathrm{d}\pi + \varepsilon\, \mathrm{KL}(\pi \mid \mu \otimes \nu) \right\}. \tag{55}$$

There are at least three common barycenter objectives.

1. The *product-reference raw entropic barycenter* minimizes $\sum_s \lambda_s \mathrm{OT}_\varepsilon(\mu, \nu_s)$. It has a shrinkage bias and can collapse.
2. The *Lebesgue/Brownian-reference Schrodinger barycenter* differs by a marginal entropy term. With $H(\mu) = \int \rho \log \rho$, the family

$$\sum_s \lambda_s \mathrm{OT}_\varepsilon(\mu, \nu_s) + \tau H(\mu)$$

has product reference at $\tau = 0$ and the Lebesgue/Brownian convention at $\tau = \varepsilon$. The latter has a blurring bias.

3. The *debiased Sinkhorn divergence*

$$S_\varepsilon(\mu,\nu) = \mathrm{OT}_\varepsilon(\mu,\nu) - \frac{1}{2}\mathrm{OT}_\varepsilon(\mu,\mu) - \frac{1}{2}\mathrm{OT}_\varepsilon(\nu,\nu) \tag{56}$$

cancels the reference-dependent marginal entropy terms and much of the entropic bias.

This taxonomy and the doubly regularized interpolation are developed in [16]. Pairwise Gaussian entropic OT has a Gaussian optimal coupling, but that pairwise statement alone does not prove that a barycenter over all probability measures is Gaussian.

10.2 Gaussian covariance equations

Let the inputs be $\mathcal{N}(m_s, A_s)$ and write $\delta = \varepsilon/2$. The mean is again $m = \sum_s \lambda_s m_s$. Within the Gaussian manifold, the product-reference raw covariance equation is

$$B = \sum_s \lambda_s (B^{1/2} A_s B^{1/2} + \delta^2 I_d)^{1/2} - \delta I_d. \tag{57}$$

More generally, adding $\tau H(\mu)$ gives the Gaussian stationarity equation

$$B = \sum_s \lambda_s (B^{1/2} A_s B^{1/2} + \delta^2 I_d)^{1/2} + (\tau - \delta) I_d. \tag{58}$$

Thus the Lebesgue/Brownian choice $\tau = \varepsilon$ changes $-\delta I_d$ into $+\delta I_d$. The product-reference equation and numerical fixed-point iteration are derived by Mallasto–Gerolin–Minh [14]; their general multivariate barycenter is Gaussian-restricted, and they leave convergence of the covariance iteration open.

For the debiased Sinkhorn divergence, Janati–Muzellec–Peyre–Cuturi prove a stronger arbitrary-dimensional result: minimization over a class of sub-Gaussian probability measures has a Gaussian minimizer. Its covariance satisfies

$$\sum_s \lambda_s (B^{1/2} A_s B^{1/2} + \delta^2 I_d)^{1/2} = (B^2 + \delta^2 I_d)^{1/2}. \tag{59}$$

See [15, Theorem 2]. An associated covariance Picard map is

$$\mathcal{T}_\varepsilon(B) := \left[\left(\sum_s \lambda_s (B^{1/2} A_s B^{1/2} + \delta^2 I_d)^{1/2} \right)^2 - \delta^2 I_d \right]^{1/2}. \tag{60}$$

The existence proof uses an invariant spectral set and Brouwer, not a contraction of (60). Uniqueness within broad Gaussian covariance classes, including infinite-dimensional versions, is studied in [17].

10.3 Scalar anti-contraction and eventual-contraction checks

For scalar variances $a_s, b > 0$, define

$$F_P(b) = \sum_s \lambda_s \sqrt{a_s b + \delta^2} - \delta, \tag{61}$$

$$F_L(b) = \sum_s \lambda_s \sqrt{a_s b + \delta^2} + \delta, \tag{62}$$

$$F_S(b) = \sqrt{\left(\sum_s \lambda_s \sqrt{a_s b + \delta^2} \right)^2 - \delta^2}. \tag{63}$$

The first two derivatives are

$$F_P'(b) = F_L'(b) = \sum_s \lambda_s \frac{a_s}{2\sqrt{a_s b + \delta^2}}, \tag{64}$$

which can exceed one near $b = 0$ for legitimate parameters. Thus entropic regularization does not automatically produce a Euclidean contraction.

For equal inputs $a_s = a$, the debiased map simplifies exactly to

$$F_S(b) = \sqrt{ab}, \tag{65}$$

independently of ε. It therefore reproduces all the scalar raw-map phenomena from theorem 7.1: non-Lipschitz behavior at the boundary, possible one-step expansion on a wide interval, and eventual contraction on each fixed invariant interval bounded away from zero. No contraction theorem uniform in the lower spectral bound can hold.

For equal isotropic inputs aI_d, the doubly regularized covariance is explicit [16]:

$$b = \frac{\left(a + \sqrt{(a-\varepsilon)^2 + 4a\tau}\right)^2 - \varepsilon^2}{4a}. \tag{66}$$

Hence product reference gives $b = (a - \varepsilon)_+$, while the Lebesgue/Brownian choice gives $b = a + \varepsilon$. The two common raw conventions have opposite biases.

10.4 Algorithms and theorem scopes

- On a fixed grid, raw entropic barycenters can be solved by matrix scaling/iterative Bregman projections; convergence follows from the KL projection framework [20, 21].
- Debiased fixed-grid Sinkhorn barycenters have efficient scaling algorithms, but empirical linear behavior should not be restated as a proof of contraction of (60) [18].
- Free-support alternatives include Frank–Wolfe and functional gradient schemes with sublinear objective or stationarity guarantees [22, 23].
- A contraction proved for an inner pairwise Gaussian dual-potential iteration is not a contraction of the barycenter covariance maps (57), (58), or (60).

The safest preservation statements are therefore:

- product-reference raw, general d: Gaussian-restricted equation;
- Lebesgue raw: full Gaussian closure is known in particular for two multivariate inputs and for multiple inputs in one dimension [19, 18];
- debiased Sinkhorn: a full sub-Gaussian closure theorem in arbitrary dimension [15].

11 Unbalanced Gaussian barycenters and WFR/HK

11.1 Ambient Hellinger–Kantorovich geometry

The dynamic Wasserstein–Fisher–Rao, also called Hellinger–Kantorovich, action combines transport and reaction. One normalization is

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = r_t \rho_t, \qquad \int_0^1 \int \left(\frac{1}{\alpha} \|v_t\|^2 + \frac{1}{\beta} r_t^2 \right) \rho_t \, \mathrm{d}x \, \mathrm{d}t, \tag{67}$$

with $\alpha, \beta > 0$. Equivalent static entropy-transport formulations use a truncated cosine cost; see [24, 25].

The term "Gaussian Hellinger–Kantorovich" is potentially misleading. In one usage, GHK refers to a quadratic-cost KL-unbalanced transport discrepancy; "Gaussian" describes the ground kernel, not closure of Gaussian measures. Liero–Mielke–Savare show that, for $g(r) = \arccos(e^{-r^2/2})$, this discrepancy is related by $\mathrm{GHK}_d = \mathrm{HK}_{g\circ d}$; ordinary HK is the induced length metric in the appropriate setting [24, Section 7.8].

11.2 An exact negative result at the Hellinger endpoint

At the pure-reaction endpoint, the square-root embedding linearizes the squared Hellinger distance. For nonnegative densities ρ_s, the unconstrained-mass barycenter is

$$\boxed{\rho_H^\star(x) = \left(\sum_s \lambda_s \sqrt{\rho_s(x)}\right)^2 .} \tag{68}$$

Proposition 11.1 (Gaussian inputs are not preserved by Hellinger barycenters)**.** *Let $\rho_s = a_s g_s$, where $a_s > 0$ and g_s is the density of $\mathcal{N}(m_s, \Sigma_s)$. Then*

$$\rho_H^\star = \sum_{s,t} \lambda_s \lambda_t \sqrt{a_s a_t} \sqrt{g_s g_t}. \tag{69}$$

Each $\sqrt{g_s g_t}$ is a positive constant times a Gaussian density with

$$\overline{\Sigma}_{st} = 2(\Sigma_s^{-1} + \Sigma_t^{-1})^{-1}, \tag{70}$$

$$\overline{m}_{st} = (\Sigma_s^{-1} + \Sigma_t^{-1})^{-1}(\Sigma_s^{-1} m_s + \Sigma_t^{-1} m_t). \tag{71}$$

Thus (69) is generically a finite Gaussian mixture, not a single Gaussian.

Proof. Minimizing $\sum_s \lambda_s \|q - \sqrt{\rho_s}\|_{L^2}^2$ over $q \geq 0$ gives $q = \sum_s \lambda_s \sqrt{\rho_s}$, proving (68). Expanding the square gives (69). Completing the square in the product of the two Gaussian half-densities gives (70) and (71). Unless all components coincide in a degenerate way, the resulting mixture is not one Gaussian. □

The normalizing constant of the cross term is the Bhattacharyya affinity

$$\int \sqrt{g_s g_t} = \frac{\det(\Sigma_s)^{1/4} \det(\Sigma_t)^{1/4}}{\det((\Sigma_s + \Sigma_t)/2)^{1/2}} \exp\left[-\frac{1}{8}(m_s - m_t)^\top \left(\frac{\Sigma_s + \Sigma_t}{2}\right)^{-1} (m_s - m_t)\right]. \tag{72}$$

11.3 Why Gelbrich's Gaussianization proof cannot transfer

The obstruction is stronger than lack of a known moment formula.

Proposition 11.2 (Moment Gaussianization is not HK-continuous)**.** *Let $\gamma = \mathcal{N}(0,1)$ on $\mathbb{R}$ and*

$$\mu_\varepsilon = (1-\varepsilon)\gamma + eps\mathcal{N}(\varepsilon^{-1/2}, 1). \tag{73}$$

Then the Hellinger and HK distances from μ_ε to γ tend to zero, but

$$\mathcal{G}(\mu_\varepsilon) = \mathcal{N}(\sqrt{\varepsilon}, 2-\varepsilon) \longrightarrow \mathcal{N}(0,2) \neq \gamma. \tag{74}$$

All covariances lie in the compact interval $[1,2]$. Hence moment Gaussianization is not continuous, and therefore not nonexpansive, for HK even under a compact covariance bound.

Proof. The L^1 distance between μ_ε and γ is at most 2ε. For any common dominating measure ξ,

$$\int \left(\sqrt{\frac{\mathrm{d}\mu_\varepsilon}{\mathrm{d}\xi}} - \sqrt{\frac{\mathrm{d}\gamma}{\mathrm{d}\xi}} \right)^2 \mathrm{d}\xi \leq \|\mu_\varepsilon - \gamma\|_{L^1} \leq 2\varepsilon.$$

HK can realize pure reaction, so it is bounded above, up to normalization, by the Hellinger distance. Thus $\mathrm{HK}(\mu_\varepsilon, \gamma) \to 0$.

The mean of (73) is $\sqrt{\varepsilon}$, its second moment is 2, and its variance is $2 - \varepsilon$, proving (74). Since HK is a metric and metrizes the relevant narrow topology for bounded masses, the distance from $\mathcal{N}(\sqrt{\varepsilon}, 2 - \varepsilon)$ to γ tends to the strictly positive distance between $\mathcal{N}(0, 2)$ and $\mathcal{N}(0, 1)$. □

The conceptual difference is that W_2 convergence includes convergence of second moments, whereas HK allows a vanishing amount of mass to travel far away or to be created/destroyed. Covariance is therefore not controlled by the ambient HK topology.

11.4 A formal geodesic obstruction for positive transport and reaction

For a smooth HK geodesic, an optimal potential satisfies, after a choice of normalization,

$$\partial_t \rho = -\alpha \nabla \cdot (\rho \nabla \phi) + \beta \rho \phi, \tag{75}$$

$$\partial_t \phi + \frac{\alpha}{2} \|\nabla \phi\|^2 + \frac{\beta}{2} \phi^2 = 0. \tag{76}$$

Suppose formally that $\rho_t = \kappa_t \mathcal{N}(m_t, \Sigma_t)$ stays Gaussian on a time interval. Then $(\partial_t \rho_t)/\rho_t$ is a quadratic polynomial in x. The weighted elliptic equation (75) has a quadratic finite-energy solution ϕ_t (this can be read from the Gaussian Hermite decomposition). If the quadratic part is nonzero, the term ϕ_t^2 in (76) has a nonzero quartic part, impossible because the other terms have degree at most two. Hence the quadratic part vanishes. If a linear part remains, ϕ_t^2 has a nonzero quadratic part while $\partial_t \phi_t$ is affine and $\|\nabla \phi_t\|^2$ is constant, again a contradiction. Therefore ϕ_t is spatially constant: only mass changes, while mean and covariance stay fixed.

This is a smooth/formal total-geodesy obstruction, not a substitute for a full weak-geodesic theorem. It explains structurally why the Wasserstein Gaussian closure mechanism disappears as soon as the reaction term $\beta \phi^2$ is present. A particular HK gradient flow may nevertheless preserve Gaussians; for example, the HK-Boltzmann flow with a Gaussian target studied in [28]. Invariance of one vector field does not imply that the Gaussian family is geodesically closed or closed under ambient barycenters.

11.5 Three legitimate unbalanced routes

1. Ambient HK/WFR barycenter. Solve in the full measure space. Existence, multimarginal formulations, and Dirac structure are studied in [26, 27]. After discretization, generalized Sinkhorn/diagonal-scaling algorithms handle entropy-regularized unbalanced transport and barycenters [25]. The result is an unrestricted measure and should not be expected to be Gaussian.

2. Gaussian-restricted BWFR geometry. Constrain paths a priori to

$$\rho = \kappa \mathcal{N}(m, \Sigma). \tag{77}$$

The projected/reduced Gaussian structure derived in [28] has a co-metric operator which, on a cotangent triple (S, u, k), takes the form

$$K^G_{\alpha,\beta}(\Sigma, m, \kappa)[S, u, k] = \left(\frac{2\alpha}{\kappa}(S\Sigma + \Sigma S) + \frac{2\beta}{\kappa}\Sigma S \Sigma, \frac{\alpha}{\kappa}u + \frac{\beta}{\kappa}\Sigma u, \beta\kappa k\right). \tag{78}$$

The induced path distance d_G is obtained by inverting this positive operator in the action. Its mean and mass terms include

$$\kappa \dot{m}^\top (\alpha I_d + \beta\Sigma)^{-1}\dot{m} + \frac{\dot{\kappa}^2}{\beta\kappa}; \tag{79}$$

the covariance term is the inverse of the Lyapunov/Riccati operator displayed in (78). Barycenters for d_G are Gaussian by construction and can be computed by Riemannian shooting, log-map gradient descent, or direct path optimization. Since the path class is restricted,

$$\mathrm{WFR}_{\mathrm{ambient}} \leq d_G \tag{80}$$

for Gaussian endpoints in compatible normalization. This is a constrained surrogate, not an ambient preservation theorem.

3. Quadratic-cost KL-unbalanced and semi-unbalanced models. Entropic quadratic unbalanced OT between scaled Gaussians has a scaled Gaussian optimal plan [15]. Recent work also gives closed Gaussian pairwise formulas without coupling entropy for quadratic-cost KL-unbalanced OT [29]; a Gaussian barycenter theorem is a separate question. Semi-unbalanced OT with one hard marginal does admit a Gaussian barycenter theorem and Bures-gradient algorithms [30]. It is asymmetric and is not the ambient HK/WFR metric, but the hard barycenter marginal restores a moment-projection mechanism and can be useful for robust averaging.

12 A research program for the raw map

The contraction question is now narrow enough to be attacked directly.

1. **Settle the local Jacobian.** Prove or disprove (54). The Sylvester representation (45), the universal radial eigenvalue 1/2, and the exact commuting spectrum (38) provide strong structure.

2. **Find the right inner product.** The general Jacobian need not be visibly self-adjoint in Frobenius geometry. A data- or fixed-point- dependent Lyapunov inner product might reveal positivity and an upper spectral bound. Any adapted norm must then be related quantitatively to the desired metric on the compact domain.

3. **Control products, not only factors.** Individual derivative norms may exceed one far from the solution. Eventual contraction requires decay of $D\Psi(\Psi^{k-1}X)\cdots D\Psi(X)$ uniformly in X. The scalar proof works precisely because early expansion is followed by exponent halving.

4. **Exploit spectral trapping.** The explicit boxes (49) forget orientation but rapidly stabilize the scale. Sharper bounds should propagate eigenvector/alignment information or use the transport maps $T_s(X)$ rather than separate min/max eigenvalues.

5. **Separate proof goals.** Global convergence of raw Picard, local linear convergence, one-step contraction, and eventual contraction on a compact are different statements. The globally convergent K update already solves the computational problem; a raw-Ψ contraction theorem would mainly clarify the nonlinear matrix equation and provide an alternate solver.

13 Final perspective

The Gaussian barycenter covariance equation is well understood as an optimality condition, and the barycenter itself is unique. What is not justified is the inference

$$\text{unique fixed point} \quad \Longrightarrow \quad \text{raw fixed-point map is a contraction.}$$

The classical algorithmic literature avoids this gap by iterating the average optimal transport map, producing $K(X) = X^{-1/2}\Psi(X)^2X^{-1/2}$ and a variance descent inequality.

For the raw map, the scalar answer is more positive than the initial suspicion: on each fixed compact interval separated from zero, a high enough iterate is a strict contraction. Yet the exact constant proves that no iterate can be chosen uniformly as the lower bound approaches the PSD boundary. The same obstruction survives in every dimension through isotropic data. In the commuting case, even arbitrary off-diagonal perturbations are locally contracted, with an explicit derivative spectrum in $[1/2, 1)$. The exact Sylvester derivative and compact-box certificate reduce the genuinely open part to noncommutative uniform control of derivative products.

Entropic regularization does not erase this issue; different entropy conventions produce different biases and covariance equations, and scalar maps still need not contract in one step. In the unbalanced setting the situation changes more radically: ambient HK/WFR does not control moments, moment Gaussianization is not continuous, and Hellinger barycenters of Gaussians are generically Gaussian mixtures. Gaussian-restricted BWFR or semi-unbalanced Gaussian models remain useful, provided their restricted or asymmetric nature is stated explicitly.

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